
Kun Qian

United States | +1-831-239-8201

kunqian.usa@gmail.com

personal website | linkedin |


## Personal Statement

I am an experienced computer scientist and researcher who has been working on a variety of AI-centric problems (NLP, Data Integration, Active Learning, Explainability, and LLMs). I have more than 30 peer reviewed publications (with 1 best demo paper award at ISWC) and 2 granted patent. I am currently a technical lead of the GenAI team at Adobe, leading a group of engineers and researchers to develop an LLM-based AI Assistant for Adobe Express Platform customers. Previously I was a technical team lead at Apple, focusing on developing automated and highly scabale open-domain knowledge extraction pipelines that improve the quality of the Apple knowledge graph with fresh and accurate facts. Before Apple, I was working at Amazon Search, focusing on improving search quality of amazon.com, and my work contributed to generating hundreds of millions in annual revenue impact. Before Amazon, I was a senior researcher at IBM Research working on research problems related to NLP, data integration, machine learning, and HCI. I also served as PC members and referees for many top computer science conferences and journals. I mentored 10+ junior researchers and interns.

## Education

### University of California, Santa Cruz — *2012-2017 / USA*

Ph.D. in Computer Science

•Advisers: Balder ten Cate, Phokion Kolaitis, Wang-Chiew Tan

•Topic: theoretical foundations of discovering data integration specifications from data examples.

### Beihang University — *2007-2010 / China*

Master of Software Engineering

•1-year exchange study at Kyushu University, Japan

### Chongqing University — *2003-2007/ China*

Bachelor of Software Engineering

## Experience

### Adobe — *2024.02 - current*

**Sr. Applied Science Manager**

•Lead a cross-functional Applied Science and Engineering team within Adobe's Agent Orchestrator group, developing state-of-the-art AI agents that empower enterprise customers to unlock data insights and streamline business operations. Spearheaded the end-to-end delivery of production-grade solutions, including text-to-SQL engines, entity resolution, and agentic conversational frameworks to solve complex, large-scale business challenges.

### Apple — *2022.06 - 2024.02*

**Team Lead and Sr. MLE**

•The technical team lead of the Open Domain Knowledge Extraction team at Apple, where I lead a team of engineers working on features that improve the richness and correctness of Apple KG that powers many Apple products. I designed and co-developed the Open-Domain Knowledge Extraction framework, an automated and scalable framework that extracts high-quality knowledge (facts) from open web and ingest into Apple KG end to end. (paper under submission)

- ML techniques: Traditional pattern-based and ML-based information extraction.
- Large-scale distributed computing framework adopted (Airflow orchestrated pyspark pipelines).
- LLM-based information extraction (zero-shot LLM prompt engineering, full parameter finetuning and parameter-efficient finetuning for open-source LLMs, e.g., vicuna).
- Built an active learning based prompt engineering framework to optimize few-shot learning for LLMs. (AAAI demo paper under submission).
- Led the intern project on fact verification that detects factual errors from text using both curated knowledge graph and open web search with zero-shot and few-shot prompted LLMs.
  - EMNLP 2023 Demo Paper Accepted (`https://arxiv.org/abs/2310.17119`).
- Mentored a summer intern on a project that significantly improved the conversion rate of the fact collection pipeline at Apple. (1 EMNLP workshop paper published in 2022)

### Amazon.com, Inc. — *2021.03 - 2022.06*

**Applied Scientist**

•I was part of the Global Search Quality team at Amazon Search. I worked on improving Amazon customers' search experiences. I led a project that brings in high-quality product substitutes to search matchset to fulfill customers' shopping intents worldwide.

- Technology used: large-scale graph algorithm, cross-encoder-based defect classifier, and graph neural network
- I have done extensive online A/B tests to verify the features before deliveries
- Feature successfully delivered to Amazon's US, DE, SG, AU, FR, IT and CA marketplaces, making positive impact to hundreds of millions of Amazon customers, increased annual revenue at 100+ million scale.

### IBM Research — *2017.02 - 2021.03*

**Research Staff Member**

•I was the technical lead for a couple of research projects on low-resource machine learning frameworks for entity matching and entity normalization. By leveraging active learning, transfer learning, and weak supervision, our frameworks can produce models that are comparable to the state-of-the-art approaches while using only 1% to 10% of the training data. A couple of systems developed by me: **PARTNER** (`https://www.youtube.com/watch?v=6DDXARJezz4`), **SystemER** (`https://www.youtube.com/watch?v=5ENye9hg-UA`).

•My research has been transferred to IBM's products including IBM Watson Health and IBM Watson NLP.

•I led a research project on explainable AI for NLP, which systematically reviewed and summarized the state-of-the-art work on this topic, and have given two well-received technical tutorials at two top AI conferences: KDD'2021 and AACL'2020. Project website: `https://xainlp.github.io/`

•I had more than 20 publications/tutorials/demos at prestigious research conferences such as AAAI, AACL, ACL, CIKM, COLING, DIS, EMNLP, ICDE, ISWC, IUI, KDD, PODS and VLDB including a best demo award at ISWC'2020. Also filed 5 patents (2 granted).

## Recent Publications

Check my full publication records at my **DBLP** and **GoogleScholar** profiles.


1. Xiou Ge, Ali Mousavi, Edouard Grave, Armand Joulin, **Kun Qian**, Ben Han, Mostafa Arefiyan, Yunyao Li
*Time Sensitive Knowledge Editing through Efficient Finetuning.*
(ACL 2024)

2. **Kun Qian**, Yisi Sang, Farima Bayat, Anton Belyy, Xianqi Chu, Yash Govind, Samira Khorshidi, Rahul Khot, Katherine Luna, Azadeh Nikfarjam, Xiaoguang Qi, Fei Wu, Xianhan Zhang, Yunyao Li
*Active Prompt Engineering: Active Learning-based Tool for Finding Informative Examples for Large Language Models.*
(Dash@NAACL 2024)

3. Farima Fatahi Bayat, **Kun Qian**, Ben Han, Yisi Sang, Anton Belyi, Samira Khorshidi, Fei Wu, Ihab Ilyas, Yunyao Li.
*FLEEK: Factual Error Detection and Correction with Evidence Retrieved from External Knowledge.*
(EMNLP 2023) *to appear*

4. Kun Qian, and other co-authors at Apple
*Growing an Industry-Scale Open Domain Knowledge Graph.*
(???? 2023) *Under Submission*

5. Shipi Dhanorkar, Marina Danilevsky, Yunyao Li, Lucian Popa, **Kun Qian**, Anbang Xu
*Explainability for Natural Language Processing. (tutorial).*
(SIGKDD 2021) The 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining.
Equal contribution, authors are listed in alphabetical order)).

6. **Kun Qian**, Marina Danilevsky, Yannis Katsis, Ban Kawas, Erick Oduor, Lucian Popa, Yunyao Li
*XNLP: A Living Survey for XAI Research in Natural Language Processing.*
(IUI 2021) Annual Conference on Intelligent User Interfaces (demo track).

7. **Kun Qian**, Poornima Chozhiyath Raman, Lucian Popa, and Yunyao Li
*Learning Structured Representations of Entity Names using Active Learning and Weak Supervision.*
(EMNLP 2020, Acceptance rate: 16.7%) The 2020 Conference on Empirical Methods in Natural Language Processing.

8. Marina Danilevsky*, **Kun Qian***, Ranit Aharonov, Yannis Katasis, Ban Kawas, Prithviraj Sen
*A Survey of the State of Explainable AI for Natural Language Processing.*
(AACL-IJCNLP 2020) The 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics.
(* Equal contribution)

9. **Kun Qian**, Poornima Chozhiyath Raman, Yunyao Li, and Lucian Popa.
*PARTNER: Human-in-the-loop Entity Name Understanding with Deep Learning.*
(AAAI-2020) The 34th AAAI Conference on Artificial Intelligence (demo).

10. **Kun Qian**, Lucian Popa, and Prithviraj Sen.
*SystemER: A Human-in-the-loop System for Explainable Entity Resolution.*
(VLDB-2019) The 45th International Conference on Very Large Data Bases.

11. Jungo Kasai, **Kun Qian**, Sairam Gurajada, Yunyao Li, Lucian Popa.
*Low-resource Deep Entity Resolution with Transfer and Active Learning.*
(ACL-2019) The 57th Annual Meeting of The Association for Computational Linguitics.

12. Phokion G. Kolaitis, Lucian Popa, and **Kun Qian**$^*$.
*Knowledge Refinement via Rule Selection*.
(AAAI-2019) The 33rd AAAI Conference on Artificial Intelligence .
(Authors are listed in alphabetical order)

13. **Kun Qian**, Nikita Bhutani, Yunyao Li, H.V. Jagadish, Mauricio Hernandez.
*LUSTRE: An Interactive System for Entity Structured Representation and Variant Generation.*
(ICDE 2018) 34th IEEE International Conference on Data Engineering.

14. **Kun Qian**, Lucian Popa, and Prithviraj Sen.
*Active Learning for Large-Scale Entity Resolution.*
(CIKM 2017) The 2017 ACM on Conference on Information and Knowledge Management.

## Patents

1. **Kun Qian**, Lucian Popa, Prithviraj Sen, and Min Li.
*Learning models for entity resolution using active learning*.
U.S. Patent 11,501,111
Status: **Granted**

2. Nikita Bhutani, Mauricio Hernandez-Sherrington, Yunyao Li, Min Li, and **Kun Qian**.
*Entity Structured Representation and Variant Generation*.
U.S. Patent 10,585,986
Status: **Granted**

## Recent Awards

**Best Demo Paper Award - ISWC 2020** *2020*
**IBM Class-A Research Accomplishment** *2017*

## Professional Affiliations and Services

**Membership**
- ACM, AAAI

**Conference Program Committee Member**
- PVLDB 2022-2023, ACL-NAACL 2022
- EMNLP 2021, DaSH@NAACL 2021, IUI 2021 (demo), NAACL 2021, AAAI 2021
- ACL 2020, IJCAI 2020, ICDE 2020 (industry), AAAI 2020
- IEEE BigData 2019
- WebDB 2018, CIKM 2018
- CIKM 2017, KDD 2017, AAAI 2017, ADAMA 2017

**Journal Referee**
- VLDBJ (2024), ACM TODS (2018, 2019), IEEE TKDE (2019)

## Skills

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|---|---|
| **Programming Languages** | Python \| Javascript/Typescript |
| **Deep Learning & Datea Science** | PyTorch \| Pytorch-Transformers \| NLTK \| Jupyter \| Matplotlib \| Numpy \| Pandas |
| **Web Programming** | Angular \| Angular Material \| W3.css \| Typescript \| Django |
| **Cloud & Distributed Computing** | AWS \| PySpark \| AirFlow |
| **Others** | IBM SystemT \| LaTeX |